

$$\lim_{n \rightarrow \infty} \left[\frac{n^2}{n+1} - \frac{n^2}{n-3} \right]$$

$$\frac{n^2(n+3) - n^2(n+1)}{(n+1)(n-3)}$$

$$(n+1)(n-3)$$

$$n^3 + 3n^2 - n^3 - n^2$$

$$= \frac{-4n^2}{(n+1)(n-3)}$$

$$(n+1)(n-3)$$

$$= \frac{-4n^2}{n^2 - 2n - 3}$$

$$= \frac{-4}{-4}$$

$$= 1$$

$$\lim_{n \rightarrow -2}$$

$$\frac{n^2 + 2n - 8}{n^2 - 4}$$

$$\frac{4 - 4 - 8}{4 - 4}$$

$$= \infty$$

$$\frac{4 + 2(-2) - 8}{(-2)^2 - 4}$$

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$$\lim_{n \rightarrow \infty}$$

$$\left(1 + \frac{2}{n} \right)^n$$

$$= \left[\left(1 + \frac{2}{n} \right)^{\frac{n}{2}} \right]^2$$

$$= \left[\left(1 + \frac{2}{2n} \right)^{\frac{2n}{2}} \right]^2$$

$$= \left[\left(1 + \frac{1}{n} \right)^n \right]^2$$

$$n = 2n$$

$$= e^2$$

$$\left(1 + \frac{2}{n} \right)^n$$

$$\left[\left(1 + \frac{2}{n} \right)^{\frac{n}{2}} \right]^2$$

$$n = 2n$$

$$\left(\left(1 + \frac{2}{2n} \right)^{\frac{2n}{2}} \right)^2$$

$$\left[\left(1 + \frac{1}{n} \right)^n \right]^2$$

$$= e^2$$

$$\lim_{x \rightarrow -3} \left(\frac{1}{x-3} - \frac{1}{|x-3|} \right)$$

$$\lim_{x \rightarrow -3}$$

$$\lim_{x \rightarrow 3^-}$$

